ESR Kit

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Westergren tubes & sodium citrate reagent

What it is used for

An ESR kit measures the erythrocyte sedimentation rate by observing how far red blood cells settle in anticoagulated blood over a specified time. Correct blood collection, anticoagulant ratio, tube positioning, and timing help produce a reliable laboratory result.

Main parts

- Westergren tube or approved ESR tube
- Westergren stand and vertical holder
- Sodium citrate reagent or approved anticoagulant system
- Graduated measurement scale
- Blood collection tube and transfer equipment
- Timer, specimen labels, and required PPE

Before you start

1. Perform the test only if trained and authorised for clinical laboratory work.
2. Confirm the test request, patient identification, specimen requirements, and the laboratory's approved ESR method.
3. Check that the Westergren tube, stand, scale, and accessories are clean, intact, and suitable for the method.
4. Verify the sodium citrate reagent identity, concentration, expiry date, and storage conditions against the approved procedure.
5. Prepare the correct blood-to-anticoagulant ratio specified by the validated method; do not estimate volumes.
6. Wear a laboratory coat, suitable gloves, and eye protection, and prepare a clean, stable work surface.

How to use

1. Identify and label the specimen using the laboratory's approved patient-identification procedure.
2. Collect the required blood specimen using the approved collection method and appropriate precautions.
3. Mix the specimen gently as required to ensure the anticoagulant is distributed evenly; avoid vigorous shaking.
4. If using sodium citrate, combine blood and reagent in the exact ratio specified by the validated method.
5. Check the specimen for clots, leakage, insufficient volume, or other rejection criteria before testing.

6. Transfer the prepared specimen into the approved Westergren tube without introducing bubbles or spilling material.
7. Place the tube vertically in its stand on a stable, level surface, away from vibration and direct sunlight.
8. Start the timer immediately after the tube is positioned, following the laboratory method's timing instructions.
9. Keep the tube undisturbed and maintain the required ambient conditions throughout the test.
10. At the specified endpoint, read the sedimentation level using the correct scale and units for the method.
11. Record the result, method, relevant specimen details, and any required quality-control observations.
12. Review results that appear inconsistent with specimen quality or laboratory acceptance criteria before reporting.
13. Dispose of used tubes, tips, and contaminated materials in the designated biohazard waste stream.
14. Clean the work area and complete the required documentation according to laboratory policy.

⚠ Safety

- Treat all blood specimens as potentially infectious and follow standard precautions.
- Wear gloves, a laboratory coat, and eye protection when handling blood or reagents.
- Never mouth-pipette; use an approved pipette or transfer device.
- Avoid splashes and aerosol generation, and clean spills using the laboratory's approved procedure.
- Dispose of sharps and biohazard waste in designated containers.
- Do not report results from clotted, incorrectly anticoagulated, leaking, or otherwise unacceptable specimens; follow the laboratory's rejection policy.
- Use only validated procedures and quality-control requirements established by the laboratory.

Care & storage

- Clean and dry the Westergren stand and reusable accessories after use according to laboratory policy.
- Check tubes, holders, and measuring scales regularly for cracks, corrosion, or readability problems.
- Store sodium citrate reagent according to its label and laboratory requirements.
- Maintain reagent lot details, expiry records, and required quality-control documentation.
- Arrange inspection or replacement of damaged equipment and report recurring measurement problems.

Common mistakes to avoid

- Using an incorrect blood-to-citrate ratio or unsuitable anticoagulant.
- Allowing the Westergren tube to tilt or vibrate during sedimentation.
- Starting or reading the test at the wrong time.

- Testing a clotted specimen or recording the result with incorrect units.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.